

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a strong foundation in Verilog, Python, and FPGA hardware design. Seeking to leverage technical expertise in ASIC design and contribute to innovative projects within a dynamic organization like Amazon Web Services that values technical proficiency and continuous learning.

EXPERIENCE

Intern

Essar Power Hazira LTD, Surat, India

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems used in boiler-actuated power generation, contributing to a project that improved operational efficiency by 15%.
- Conducted research on link mechanisms designed to transport waste loads from boilers, enhancing system reliability and reducing downtime by 10%.
- Monitored and recorded gauge and probe data while ensuring safety compliance through seminar participation, contributing to a 100% safety record during the internship.

Intern

3D Cubic, Surat, India

Aug 2021 - Oct 2021

- Completed an internship focused on understanding and operating 3D printing technology, resulting in the successful design and fabrication of a 2 cm x 2 cm x 2 cm polyamide cube, demonstrating proficiency in rapid prototyping.
- Showcased hands-on expertise by utilizing CAD software to design components, reducing production time by 20% through efficient design practices.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, SystemVerilog, FPGA Design, VLSI Engineering, RTL Design, Assertions, Scripting (TCL, Perl), Signal Integrity, Timing Analysis, Debugging, Data Path Design, AXI Protocol, MATLAB, Cadence, Synopsys VCS, Primitime, Quartus II IDE, Proteus, P-spice, Simplify3D

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Designed SRAM with block and state diagrams for PCI to SRAM transactions, achieving a data transfer rate of 1.5 Gbps.
- Prepared timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring compliance with industry standards.

Vending Machine Simulation

Aug 2023 - Dec 2023

- Integrated finite state machines including idle, selection, payment, dispensing, and reset states, improving simulation accuracy by 30%.
- Tested and verified the design using Verilog testbenches and simulation environments, achieving a 95% success rate in hardware validation on an Intel DE-10 Lite FPGA board.

6-Bit Segmented DAC

Jan 2024 - May 2024

- Achieved expected results for both theoretical and simulated values of 12V, reducing glitch noise by 25% through effective segment implementation.
- Conducted extensive testing and validation to ensure design robustness, contributing to a 15% improvement in signal fidelity.

Static Timing Analysis for Microprocessor Netlist

Aug 2024 - Dec 2024

- Identified timing violations using detailed reports and optimized the generated slack, improving overall design performance by 20%.
- Collaborated with cross-functional teams to enhance design quality and ensure timely project delivery.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented Verilog module and Perl script to verify functionality, achieving a 100% accuracy rate in test results.
- Generated timing reports by applying design constraints, facilitating efficient static timing analysis and reducing analysis time by 30%.